SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

Code of Conduct:

I Praneeth.R (Name / Reg No) *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5
Understanding of Scenario	20%	1	1	1	1	1
Identification of Key Issues	20%	1	1	1	1	1
Application of Concepts	25%	1	1	1	1	1
Decision Making	20%	1	1	1	1	1
Clarity of Response	15%	1	1	1	1	1

25/30

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

Quality Assurance → **/25**

Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3.A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the **/64** prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.

2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Scenario	20%	Demonstrates clear and complete understandin	Shows good understandi ng with minor gaps	Partial understandin g; misses key aspects	Misinterpret s or fails to understand the scenario

		g of the scenario and context			
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

SIMATS ENGINEERING

ASSESSMENT TOOL 2 – COMPUTER NETWORKS

Course Code: CSA07

Name: Pranesh.R

Register Number: 1912260397

CO2: Configure IP addressing, subnetting and basic routing.

Q1. Smart Campus Network – 192.168.100.0/24

We allocate the subnets one after another without overlapping. The given network has addresses from 192.168.100.0 to 192.168.100.255.

Department	Network address	Broadcast address	Valid host range	Usable hosts
School of Computing (/25)	192.168.100.0	192.168.100.127	192.168.100.1 – .126	126
School of Electronics (/26)	192.168.100.128	192.168.100.191	192.168.100.129 – .190	62
School of Mechanical (/27)	192.168.100.192	192.168.100.223	192.168.100.193 – .222	30

Remaining unused range: 192.168.100.224 – 192.168.100.255, which is 192.168.100.224/27.

Reason: /25 has 128 addresses, /26 has 64 addresses, and /27 has 32 addresses. The usable hosts are total addresses minus network and broadcast addresses.

Q2. Company Network – 10.10.0.0/16

Assuming the subnets are allocated sequentially from the beginning of the /16 block:

Department	Network address	Broadcast address	Valid host range	Usable hosts
Software Development (/24)	10.10.0.0	10.10.0.255	10.10.0.1 – 10.10.0.254	254
Quality Assurance (/25)	10.10.1.0	10.10.1.127	10.10.1.1 – 10.10.1.126	126
Human Resources (/26)	10.10.1.128	10.10.1.191	10.10.1.129 – 10.10.1.190	62
Executive Management (/27)	10.10.1.192	10.10.1.223	10.10.1.193 – 10.10.1.222	30

Remaining unused address range: 10.10.1.224 – 10.10.255.255 (within 10.10.0.0/16).

The unused range assumes the subnets are placed consecutively as shown. Each subnet has two addresses that cannot normally be assigned to hosts: the network address and broadcast address.

Q3. IPv6 Network – 2001:0db8:abcd:0000:0000:0000:0000:0000/64

1. Compressed notation: 2001:db8:abcd::/64

2. Full expanded notation: 2001:0db8:abcd:0000:0000:0000:0000:0000/64

3. The /64 network prefix is the first 64 bits, represented as 2001:0db8:abcd:0000::/64.

4. A /64 IPv6 subnet has 2^(128 − 64) = 2^64 addresses.

5. Total possible addresses: 18,446,744,073,709,551,616 (2^64). The question repeats the total-address calculation, so the answer is the same.

Note: In IPv6, there is no IPv4-style broadcast address; IPv6 uses multicast instead.

Q4. Routing Table – Destination 192.168.2.45

Given routes: 192.168.1.0/24, 192.168.2.0/24 and 192.168.3.0/24.

1. Destination subnet: 192.168.2.0/24, because 192.168.2.45 is within the range 192.168.2.1 to 192.168.2.254.

2. Matching routing entry: 192.168.2.0/24.

3. Next hop/interface: The question does not provide next-hop addresses or interface names, so the exact next hop cannot be identified. The router forwards the packet using the next-hop or outgoing interface configured for 192.168.2.0/24.

4. If no listed route matches, the router can use a default route, written as 0.0.0.0/0, pointing to an upstream router/interface. The exact next hop is not specified in the question.

Q5. Administration LAN and Finance LAN

LAN	Network address	Broadcast address	Valid host range
Administration (/26)	192.168.10.0	192.168.10.63	192.168.10.1 – 192.168.10.62
Finance (/26)	192.168.10.64	192.168.10.127	192.168.10.65 – 192.168.10.126

Three suitable computer IP addresses:

• Administration: 192.168.10.2, 192.168.10.3 and 192.168.10.4. Default gateway: 192.168.10.1.

• Finance: 192.168.10.66, 192.168.10.67 and 192.168.10.68. Default gateway: 192.168.10.65.

The gateway must belong to the same subnet as the computers. The given gateway addresses are valid host addresses in their respective /26 subnets.

Short Conclusion

Subnetting divides a large network into smaller networks. Correct network addresses, broadcast addresses, valid host ranges and gateways help devices communicate properly. Routing tables help routers forward packets toward the correct destination.